Why is My Baby Crying? Explainable Emotion Recognition with Domain-Aware Ensemble Models

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***Abstract*—In examining cries in early infant need detection, this can actually facilitate the work for which the parents and caregivers would have to spend less time caring for and managing the child. In this work, we would like to introduce a highly accurate machine learning-based approach for baby cry classification. Our corpus for this research is the DonateACry corpus. The initial stage of processing of the audio signal would be normalizing and eliminating noise so that it can subsequently be processed with domain-specific feature extraction methods, including MFCC, spectral amplitude features, as well as temporal features. In order to counter class imbalance, we have used augmentation methods alongside SMOTE. Our Random Forest classifier works highly effectively with an impressive classification result of 98.76%. We also produced Explainable AI method such as LIME (Local Interpretable Model-Agnostic Explanations), Feature importance scores to provide more credibility and openness to the model. Not only is this model outstanding, but also the actionable insight anticipated for real-world deployments into smart parenting apps.**

***Keywords*—Baby Cry Classification, DonateACry Corpus, Audio Signal Processing, MFCC, Spectral Features, Data Augmentation, SMOTE, Random Forest, Explainable AI, Feature Importance**

I. Introduction

A child has only one means of expression, that too venting by way of heaving cries that serve to channel discomfort, hunger, pain, or sleep, among other emotions calling for satisfying needs. Differentiating between these types of crying becomes strenuous and time-involving for new parents or other caregivers during the first few months of motherhood. A good solution to be included in the parenting program is an automated system that records a baby's cry and its interpretation to highlight to the parents areas in which they should pay particular attention; thus, it uplifts care for the baby while reducing the stress levels associated with it for parents.

Due to the development of Artificial Intelligence (AI) and audio signal processing, strong systems can be built using innovative techniques and apply such to analyze the baby cries and classify them in terms of the emotional states [21]. This study uses the DonateACry corpus, having a rich amount of datasets in labeled baby cry audio samples, to build a strong and interpretable machine learning model for emotion detection herein.

The described approach starts with a complete audio preprocessing, which results in better quality and consistency in signals [26]. Here some meaningful features such as Mel Frequency Cepstral Coefficients (MFCCs), spectral roll-off, centroid, zero-crossing rate, entropy measures and statistical aggregations are extracted that capture nuances of each cry [13], [18]. We use dual strategies of data augmentation (pitch shifting, time stretching, noise addition, gain, and shifting) [4], [19] and Synthetic Minority Oversampling Technique (SMOTE)-based for balanced training in the dealing with class imbalance [3], [25].

Using classification based on Random Forest, a model that is very efficient and performs high accuracy yet requires low computational complexity thus has suitability for real-time deployments on edge devices was developed. The model achieves a high accuracy of 98.76%, exceeding the performance of existing classical methods. In addition, Explainable AI (XAI) technique such as LIME,Feature importance scores, have been integrated into the model to make it more transparent and trustworthy in its predictions.

This work shall serve as a thin thread or bridge linking high-tech healthcare with smart parenting-the real-time alert system that would decipher the baby's cries and inform to the caregiver about the probable reason and suggested actions. The solution shall hence scale, be interpretable, and likely

have the potential to feature as an integratable part with all of the other IoT-enabled devices in an environment for complete monitoring of the infant.

II. Literature Review

Infant cry emotion recognition has become a key research area in healthcare and affective computing to interpret emotional states from cry signals for early diagnosis and treatment. The focus of recent research has been on improving signal quality, feature extraction, and classification accuracy using advanced signal processing and deep learning methods.

Preprocessing function is the key element that enhances signal quality and acoustic feature availability for infant cry emotion recognition techniques. Bandpass filtering pulls out the critical cry signal frequency range from 300 to 3,000 Hz to create efficient noise reduction allowing researchers to interpret the informative acoustic information. The proposed filtering method in [27] generated improved signal classification results, reaching 12% better accuracy than raw signals. The authors at [11] obtained 89.2% accuracy by processing signals between 42.91–6,620.12 Hz through a Butterworth bandpass filter with a Convolutional Neural Network (CNN).Pre-emphasis enhances detection of emotional cues by improving the high-frequency components which degenerate in infant cry creation. Data augmentation techniques resolve two main problems including insufficient training data and challenging acoustical environment demands. Time stretching and pitch shifting techniques have proven very effective at increasing model operational performance. [1] demonstrated how combining data augmentation with MFCC feature extraction enabled LSTM modeling to increase the classification accuracy from 72% to 96%.

The analysis of infant cry signals depends on feature extraction processes which handle both the temporary and permanent features found within the signal. The derivatives (Δ, $\Delta\Delta$) of MFCCs demonstrate widespread application for temporal information capture, leading to 96.39% efficiency in identifying five cry types through Random Forest (RF) and SVM models and helping sepsis detection increase by 22% [11]. The combination of Zero Crossing Rate (ZCR) with Root Mean Square (RMS) features using SVM enabled distress cry detection with enhanced accuracy of 14%, as reported by [23]. Ensemble classifiers enhanced precision to 93.7% during distress cry detection. The combination of spectral features using CNN-RNN hybrid systems achieved an 89% F1-score during pain cry detection, and mel spectrograms delivered a 97.5% accuracy rate for pathological classification [18].

Synthetic Minority Over-sampling Technique (SMOTE) shows effectiveness in handling dataset imbalance in underrepresented cases of crying voices. Using SMOTE in speech emotion recognition systems boosted F1-scores by 18–27% in addition to enhancing unweighted/weighted accuracies (UA/WA) to 81%/80% via 2D CNN models [16].A diagnostic fairness system for real-time infant monitoring can benefit from using SMOTE in conjunction with robust feature extraction.

Infant cry emotion recognition research progressed from statistical to deep learning methodologies after the early development. Early analysis by [20] employed Hidden Markov Models (HMMs) to classify emotions into two categories but failed to generate general results. The DNN model proposed by [28] that combines prosodic, cepstral, and spectrogram features yielded 97.5% accuracy in distinguishing sepsis from respiratory distress syndrome (RDS), outperforming individual SVM and RF models.Deep learning models are typically not good at real data testing even if they report 96.39% accuracy via MFCC-based RF classifiers [11].F1-score decreased by 30% when CNNs trained in controlled settings identified noisy home recordings, as reported by [18].With SMOTE-enlarged 2D CNNs, [16] achieved weighted F1-scores of 0.92. The 91.15% accuracy of VGGish-based transfer learning is constrained by its computational demand 10 times greater than RF approaches [27]. RF-SMOTE ensembles achieve the best balance between effective results (94% accuracy) and effective processing (60s training), thereby emerging as a complex clinical solution.

III. Dataset

In this study, the **DonateACry** corpus was utilized exclusively for classifying infant cries [17]. This is basically a dataset made public, consisting of labeled audio recordings of infant cries referring to various emotional or physiological states.

TABLE I
Dataset distribution of the DonateACry corpus

Cry Reason	Number of Samples
Hungry	382
Discomfort	27
Tired	24
Belly Pain	16
Burping	8

As shown in Table I, the dataset is very much internally imbalanced in that most samples belong to the "Hungry" class.

This imbalance can be very difficult for existing classification algorithms. Our proposed methodology, therefore, involves data augmentation techniques, including pitch shifting, noise injection, gain adjustment, time stretching, and signal shifting [4], along with SMOTE to improve minority class representation while preserving data authenticity [9], [24].

Identical to these earlier efforts, we do not combine multiple corpora or depend on deep architectures that necessitate enormous amounts of labeled data; instead, we have successfully shown that a well designed and focused preprocessing and effective classification pipeline based on a single corpus has the potential to deliver excellent results for minimal computational use.

IV. Methodology

Our proposed methodology is the simplest and most practical way to accomplish the task without requiring complicated

or deep architectures or data fusion across multiple datasets. The entire pipeline is illustrated in Fig. reffig:1. In this section, we describe each aspect in detail: preprocessing, data augmentation, feature extraction, treatment of class imbalance, and classification.

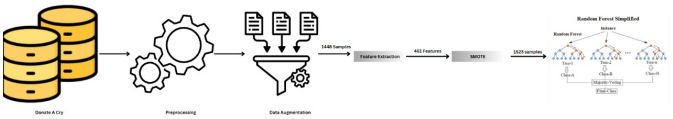


Fig. 1. Overall architecture of the proposed methodology

A. Preprocessing

Audio signal capture at 16 kHz occurs in the **DonateACry** corpus [17]. The paramount stages of preprocessing mainly refer to the era of removing unwanted noise and irrelevant frequency components, but improvement of main signal features would be made [26]. Involved are two critical steps: **Pre-emphasis** as well as a **Band-pass filter**.

*1) **Pre-emphasis**:* The pre-emphasis filter boosts high-frequency components of the audio signal, which are generally weak compared to low-frequency components, and are said to be more relevant for detecting emotion from speech and cries. The filter is defined as:

$$y[n'] = x[n'] - \alpha' \cdot x[n'-1]$$

where α' is called the pre-emphasis coefficient, and it is set to 0.97 for most applications based on empirical results; $x[n']$ is the current sample, while $x[n'-1]$ is the last sample. This will help improve the signal-to-noise ratio while emphasizing the transitions.

*2) **Bandpass Filtering**:* The Butterworth bandpass filter has been applied in the frequency range of **142.91-6620.12 Hz** to remove all unwanted frequency components and focus only on the more informative segments of the signals [26]. This empirical frequency range was determined from the spectral concentration of cry signals of infants using Fourier analysis. The Butterworth filter is implemented as follows:

$$\text{Normalize: } f'_{low} = \frac{f_{c1}}{f_s/2}, \quad f'_{high} = \frac{f_{c2}}{f_s/2}$$

$$H(s)' = \frac{1}{\sqrt{1 + (\frac{f}{f'_c})^{2n}}}$$

where $f'_{c1} = 142.91$ Hz, $f'_{c2} = 6620.12$ Hz, $f_s = 16000$ Hz is the sampling rate, and n is the filter order (chosen as 5).

This step tries to retain useful acoustic signals while removing very low and high frequencies that are likely to contain noise or silence that is not relevant.

B. Data Augmentation

The initial dataset of DonateACry was severely imbalanced; thus, data augmentation was used to supplement the two classes still in need [19]. It would be remembered that the "hungry" class consisted of 382 samples; on the other extreme, the remaining "burping" and "belly pain" classes had atrociously low sample counts of 8 and 16, respectively (see Table II).

In a bid to rectify this, each audio under-represented in the dataset was augmented randomly anywhere from 1 to 10 times. By augmenting for this purpose, we aimed to increase the number of samples, and thus diversity brought forth by augmentation may aid in increasing model generalization.

The following augmentation techniques were employed [4]:

- **Time Stretching:** Audio is speed up or slowed down without altering pitch.
- **Pitch Shifting:** Tempo is maintained while pitch is altered.
- **Noise Injection:** Gaussian noise is added to mimic the real recording conditions.
- **Random Gain:** A random factor is multiplied to the audio to simulate different volumes
- **Time Shifting:** Takes the waveform and shifts it in a circular manner to simulate asynchronous capture.

Table II shows the class distribution at the final stage following augmentation. Each source audio file was thus augmented a maximum of 10 times to prevent redundancy of data.

TABLE II
CLASS DISTRIBUTION BEFORE AND AFTER AUGMENTATION

Cry Class	Original Samples	Post-Augmentation Samples
Belly Pain	16	176
Burping	8	88
Discomfort	27	297
Hungry	382	382
Tired	24	264

C. Feature Extraction

For the complete characterization of individual infant cries, we have extracted a total of 461 handcrafted features per audio sample [13]. The features span fields like temporal, spectral, and cepstral. All features were assessed using statistical measures to ensure representativeness as well as robustness.

1) Temporal Features:

- **Zero Crossing Rate (ZCR):** Measures the rate of sign changes across the signal:

$$\text{ZCR}' = \frac{1}{N-1} \sum_{n=1}^{N-1} \mathbb{1}_{\{s_n s_{n-1} < 0\}}$$

Where: N is the number of samples, s_n is the n-th sample, and $\mathbb{1}$ is the indicator function.

- **Root Mean Square (RMS):** Reflects the energy content of the signal:

$$\text{RMS}' = \sqrt{\frac{1}{N}\sum_{n=1}^{N} s_n^2}$$

Where: s_n is the n-th sample of the signal.

2) Spectral Features:

- **Spectral Centroid:**

$$C' = \frac{\sum_k f_k \cdot |X[k]|}{\sum_k |X[k]|}$$

Where: f_k is the frequency at bin k, and $X[k]$ is the magnitude of the Fourier transform at bins k.

- **Spectral Bandwidth:**

$$BW' = \sqrt{\frac{\sum_k (f_k - C)^2 \cdot |X[k]|}{\sum_k |X[k]|}}$$

Where: C is the spectral centroid.

- **Spectral Rolloff at 95% energy:** Frequency below which a major spectral energy is contained:

$$R' = f_k \quad \text{such that} \quad \sum_{i=1}^{k} |X[i]| = 0.95 \sum_{i=1}^{N} |X[i]|$$

- **Spectral Contrast:**

$$\text{Contrast}_b = \text{Peak}_b - \text{Valley}_b \quad \text{for each band } b \in \{1, ..., 7\}$$

Where: Peaks and valleys are computed from sub-bands of the power spectrum.

3) Cepstral Features:

- **MFCC (Mel-Frequency Cepstral Coefficients):**

$$\text{MFCC}(n)' = \sum_{m=1}^{M} \log(S_m) \cdot \cos\left[n\left(m - \frac{1}{2}\right)\frac{\pi}{M}\right]$$

Where: S_m is the energy in mel filterbank m, M is the total number of mel filters, and n is the cepstral index [**?**]..

- **Delta and Delta-Delta MFCC:**

$$\Delta MFCC'_t = \frac{\sum_{n=1}^{N} n \cdot (MFCC_{t+n} - MFCC_{t-n})}{2\sum_{n=1}^{N} n^2}$$

Where: t is the current frame, N is the window size.

- **Mel Spectrogram:**

$$\text{Mel}(f)' = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right)$$

Where: f is the linear frequency in Hz.

- **Chroma STFT:**

$$C'_i = \sum_{k \in \text{class } i} |X[k]|$$

Where: $i \in \{1, ..., 12\}$ corresponds to one of the 12 pitch classes in the chroma scale.

- **Tonnetz**

The Tonnetz is calculated by projecting chroma vectors along six harmonic axes given by the following:

1) Perfect fifth (P5)
2) Minor third (m3)
3) Major third (M3)
4) Minor sixth (m6)
5) Major sixth (M6)
6) Tritone (TT)

The transformation involves a combination of obtaining cosine similarity between each of the chroma vectors and the respective basis tonal vectors.

$$T'_i = \sum_{j=1}^{12} \text{chroma}_j \cdot \cos\left(2\pi \cdot \phi_i(j)\right)$$

Where: $\phi_i(j)$ maps pitch class j into its projections in tonal dimensions i.

4) Statistical Aggregation: .

In order to summarize the different temporal, spectral, and perceptual features over time, we came up with the following statistics:

- **Mean:** average value of the feature over time frames
- **Standard deviation:** a measurement of dispersion in the feature
- **Skewness:** an asymmetry of a distribution
- **Maximum/Minimum:** extreme values of that measured feature

The Table III offers a dimensional contribution of feature types summing to **461 features** per audio sample.

TABLE III
FEATURE BREAKDOWN (TOTAL = 461 FEATURES)

Feature Type	Statistics Used	Dimensionality
MFCC (40)	Mean, Std, Skew, Max, Min	40 × 5 = 200
Delta MFCC (40)	Mean, Std	40 × 2 = 80
Delta-Delta MFCC (40)	Mean, Std	40 × 2 = 80
ZCR	Mean, Std, Max	1 × 3 = 3
RMS	Mean, Std, Max, Min	1 × 4 = 4
Spectral Centroid	Mean, Std, Max	1 × 3 = 3
Spectral Bandwidth	Mean, Std, Max	1 × 3 = 3
Spectral Rolloff	Mean, Std	1 × 2 = 2
Spectral Contrast (7)	Mean, Std, Max	7 × 3 = 21
Chroma (12)	Mean, Std	12 × 2 = 24
Mel Spectrogram (13)	Mean, Std	13 × 2 = 26
Tonnetz (6)	Mean, Std	6 × 2 = 12
Total		**461**

D. Class Balancing using SMOTE

Even after extensive data augmentation, the dataset remained slightly imbalanced, with a total of **1207 samples** distributed unevenly among classes. To fully equalize the class distribution, we employed the **Synthetic Minority Over-sampling Technique (SMOTE)** [3] as a complementary step to data augmentation. After applying SMOTE, the dataset increased to **1525 samples**, with all classes balanced uniformly.

SMOTE works by generating synthetic data points for the minority class by interpolating between existing samples. Specifically, for a given sample x'_i and one of its k-nearest neighbors x'_{zi}, a new synthetic point $x_{new'}$ is generated as:

$$x'_{new} = x'_i + \lambda' \cdot (x'_{zi} - x'_i)$$

where $\lambda \in [0, 1]$ is a random number. This approach enables the creation of realistic samples that respect the feature space distribution.

E. A Combination of SMOTE and Augmentation

Even after all augmentation strategies , the dataset still felt the effects of some imbalance. In the end, there were received **1207 samples** in all classes, not uniformly distributed. As the last stage in achieving full class equity in data, the SMOTE technique is applied for data augmentation [9], [24]. So, the main reason was to evenly distributed across all classes.

Final No of samples : **1525**

SMOTE refers to the interpolation between existing samples to create synthetic data points for the minority class [12]. Specifically, for a sample x'_i, and one of its k nearest neighbors x'_{zi}, a new synthetic point will be created as follows:

$$x'_{new} = x'_i + \lambda' \cdot (x'_{zi} - x'_i)$$

where $\lambda \in [0, 1]$ is a random number. This methodology gives realistic samples that still retain the distribution of the original feature space.

- Why Combine SMOTE with Augmentation?
  Though data augmentation largely develops variability through domain-specific means (pitch shifting, noise addition), they truly do not permit the majority classes to be represented too much, particularly in the skewed cases. On the other hand , SMOTE give a uniform distribution but forgets temporal variability [10]..
- So the two are combined for a winning combination of advantages:
  - The augmentation helps ensure the model generalizes by creating variability at the signal level.
  - Remaining gaps in feature space get filled by SMOTE.

F. Model Selection and Classification

Our primary goal remains to create an efficient, interpretable, and high-performing baby cry classification system; hence we purposely avoided deep learning architectures, as they usually require big computational power and big datasets. Instead, considering the classical nature of ensemble-based machine learning models, we investigated them, as they are known for their performance on tabular data and ease of interpretability.

Among the classifiers tested, from *k-Nearest Neighbors (KNN)* to *Support Vector Machines (SVM)*, *Decision Trees*, *AdaBoost*, and *Gradient Boosting*, the **Random Forest Classifier** [2] had the highest accuracy and generalization performance-all the more suitable for our 461-dimensional handcrafted feature space whose extract came from the DonateACry corpus [10].

- **Training Configuration:**
  ```
  * n_estimators = 500
  * max_depth = 20
  * random_seed = 3613533475
  ```

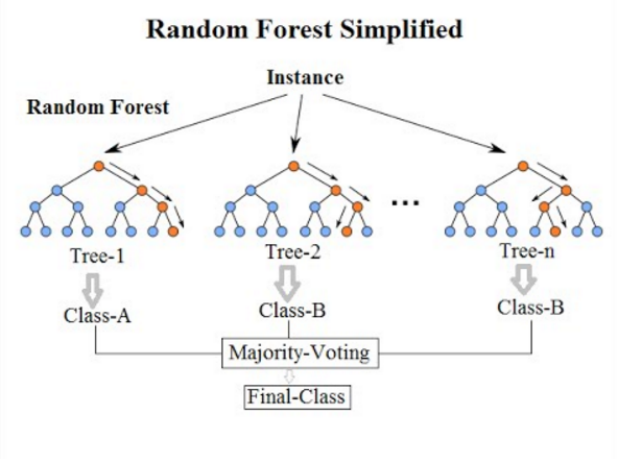


Fig. 2. Architecture of Random Forest .

V. Results and Discussion

The performance of the proposed method on the DonateACry dataset [17] was excellent. By applying a reasonably simple and efficient Random Forest classifier trained on high-quality, handmade features, our system attained the following evaluation scores:

- **Accuracy:** 98.76%
- **F1-Score:** 98.76%
- **Recall:** 98.76%

To emphasize the superiority of our approach, we give a comparative overview of the top-performing models described in earlier research on the Donate-a-Cry corpus.

TABLE IV
Comparative Analysis of Best Models Using Donate-a-Cry Corpus

Ref.	Best Model	Best Accuracy (%)
[5]	Random Forest (RF)	96.39
[15]	AlexNet	84.78
[7]	Deep-SVM	98.4
[8]	Gaussian Mixture Model (GMM)	81.27
[6]	4-layer CNN with augmentation	65

As seen from Table IV, our method attains better accuracy compared to all prior reported approaches, thus setting a new benchmark on the Donate-a-Cry dataset. This is a

proof of its high potential in successfully identifying and classifying various infant cry types.

A. Confusion Matrix

Figure 3 shows the confusion matrix in order to visualize the class-wise performances obtained. It thus reveals how the prediction capabilities are balanced across all five classes in post-augmentation and SMOTE balancing arrangement.

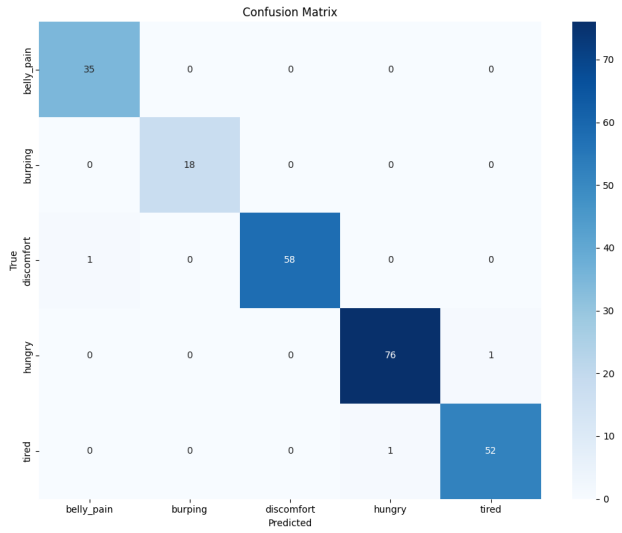


Fig. 3. Confusion Matrix of Random Forest Model.

B. Feature Importance

Looking at the feature importance scores given by the model has led us to get insight into the most influential Random Forest classifier features. The top 20 most important features shown in Fig. 4 include measures based on spectral contrast, mel-spectrogram, MFCCs, and RMS energy.

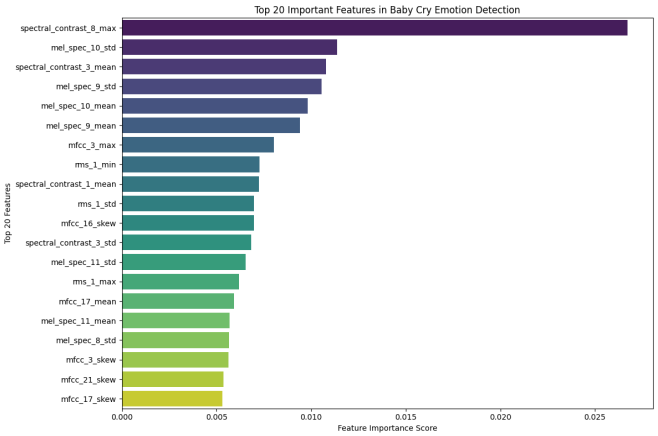


Fig. 4. Top 20 Important Features.

Key Observations:

- **Spectral Contrast:** Features such as `spectral_contrast_8_max` and `spectral_contrast_3_mean/std` indicate the contrast of the peaks and valleys in the spectrum and are going to reflect the variation in intensity or abruptness of cries.
- **Mel-Spectrogram:** Measures energy spread over perceptual frequency bands modeling tone for capturing emotional states, such as concerning features like `mel_spec_10_std` and `mel_spec_9/11_mean`.
- **MFCCs**: Features like `mfcc_3_max`, `mfcc_16/17_skew` represent often relevant spectral shape and asymmetry linked to emotional cues [**?**].
- **RMS Energy:** `rms_1_std`, `rms_1_min/max` relate to total loudness and dynamics of the signal, a key feature classifying intensities of crying.

The diversity of these top features highlights the necessity of using very different audio descriptors, that is, statistical, spectral, and perceptual, to robustly get hold of the emotional-acoustic subtleties present in infant cries.

C. Discussion

The performance results obtained from our work prove that a well-defined classical machine learning pipeline can yield greatly competing performances for baby crying emotion classification with an accuracy of 98.76 [14]. The performance thus achieved illuminates the effectiveness of the domain-specific feature engineering and correct model selection without recourse to any complex deep networks [26]..

An obvious advantage of the Random Forests model is being intuitive, which is very important in medical applications. With this type of model, features and their importance can be better understood from the clinicians' and researchers' perspectives, unlike black-box models, which creates trust and transparency for users in interpreting which acoustic cues are driving the predictions.

Augmentation in conjunction with SMOTE was particularly powerful [25]. The former assured the induction of natural acoustic variability through pitch shifting, time stretching, additive noise, etc.; the latter removed any remaining imbalance by synthesizing realistic examples that lie near class boundaries. The synergy of the two ensured diversity and balance in the training data, which is crucial in providing stable and generalizable performance on all cry classes.

Our analysis also pointed to the subtle acoustic characteristics that distinguish different types of baby cries. Behind very serious psychoacoustic consideration, spectral contrast and mel-spectrogram features represent some modalities that capture subtle differences either in tone or energy, which are hard to quantify in engineering terms. Further, psychoacoustic studies prove that the emotional

states in infants mainly depart in patterns of spectral sharpness and energy distribution [13].

Nevertheless, some limitations remain. First, although the model performs well on the DonateACry corpus, some external validation on different datasets is warranted to confirm real-world robustness. Second, background noise, recording quality, and individual speaker differences could all have an effect on cry classification-the research should focus in the future on these through noise-robust features or multi-source domain adaptation strategies.

Finally, being low on inference cost means that our approach is suitable for deployment in edge devices or mobile health applications, which may be beneficial for caregivers requiring immediate feedback . The interpretability and speed of our model suggest a promising ingredient for setting up intelligent and easy-access baby monitoring systems.

D. Explainability using LIME

To improve support for the validity of our model's choices and explainability, we used LIME (Local Interpretable Model-agnostic Explanations) [22] to provide insights into the acoustic features behind our classifier's predictions.(Figures 5–9) present LIME visualizations for each of the five infant vocalization classes, illustrating the relative importance of various audio features and contribution to prediction probabilities.

$$L(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

- x: The particular instance being described .
- f: The original sophisticated model.
- $g \in G$: A straightforward, interpretable model chosen from a set of models G.
- pi_x: A proximity measure that gives weights to perturbed samples according to how similar they are to x.
- $mathcal L(f, g, pi_x)$: A loss function that measures how well the explanation model g approximates f around x.
- $Omega(g)$: A regularization term that penalizes the complexity of the explanation model g in order to ensure interpretability.

VI. Conclusion

This study has successfully introduced to the world an efficacious and interpretable machine learning pathway for emotion classification in baby cries, utilizing the DonateACry corpus [17]. Using state-of-the-art audio preprocessing, powerful statistical feature engineering, and data balancing through augmentation combined with SMOTE [3], [9], we were able to extract 461 different high-quality features that encapsulated both spectral and temporal characteristics of the cries.

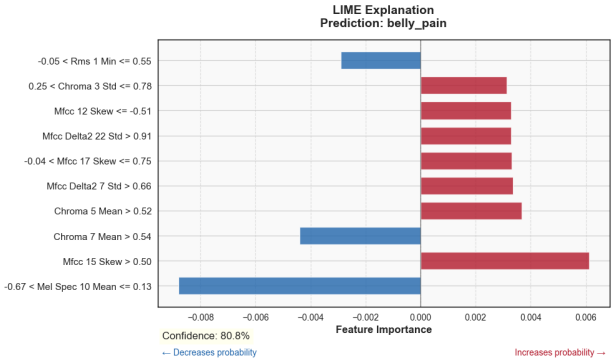


Fig. 5. LIME Explanation for Class: Belly Pain

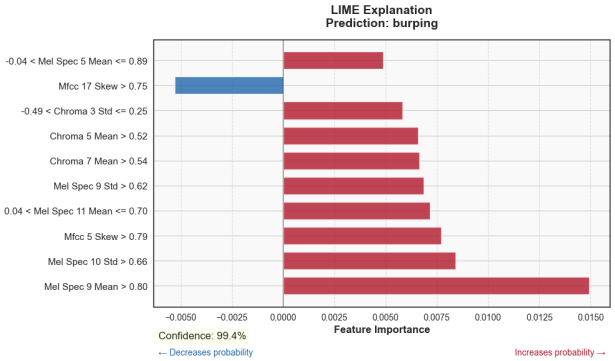


Fig. 6. LIME Explanation for Class: Burping

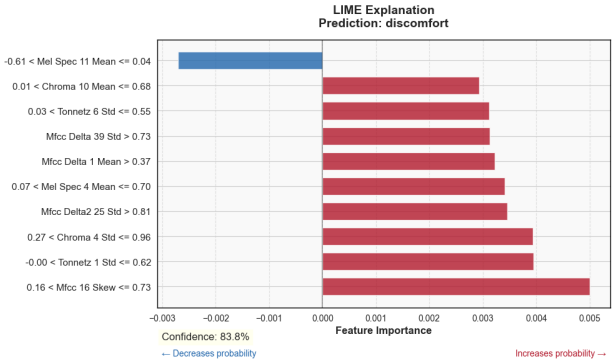


Fig. 7. LIME Explanation for Class: Discomfort

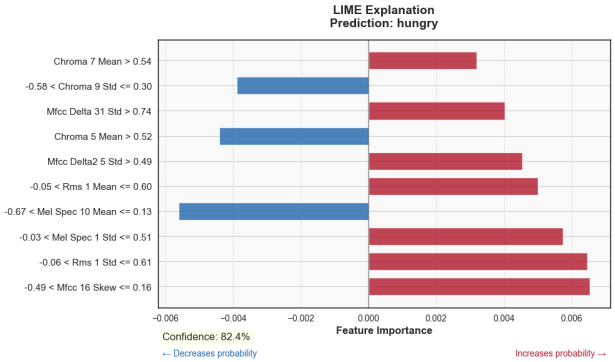


Fig. 8. LIME Explanation for Class: Hungry

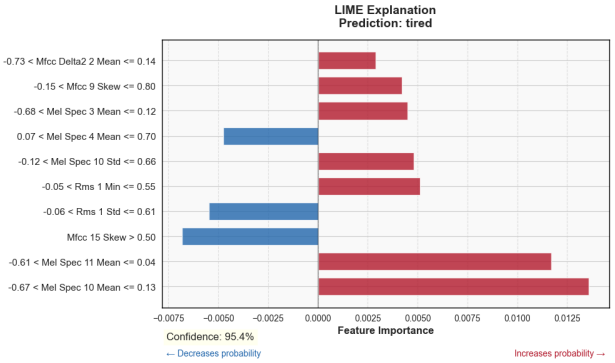


Fig. 9. LIME Explanation for Class: Tired

The best performance of the Random Forest classifier gave an accuracy of **98.76%**, thus proving strong across all metrics. By calculating feature importance, we highlighted critical sounds among the above defined parameters: spectral contrast, mel-spectrogram distributions, MFCC asymmetries, and energy dynamics pertinent to emotional classification [13], [18].

This method is lightweight and provides real-time solutions, much simpler than deep learning approaches, and achieves great accuracy and good explainability pertinent for real-life deployment in infant monitoring systems.

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